WATER SAFETY MONITORING USING IOT AND REGRESSION MODELING



Internet of Things (IoT)

IoT connects everyday devices, enabling data sharing and automated control of systems like smart cities.

Objectives

- Real-Time Tracking: Use smart sensors to constantly check water pH, solids (TDS), and temperature.
- Smart Predictions: Build a model that uses live data to "guess" if water is becoming unsafe before it actually happens.
- Calculating Risk: Create a simple "Risk Score" so anyone can understand the safety level of their water at a glance.
- Stopping Outbreaks: Catch dangerous water conditions early to prevent the spread of diseases and send out quick health alerts.
- 24/7 Remote Watch: Move away from slow, manual testing by using an automated dashboard that monitors water quality around the clock.

Motivation

- With the rapid growth of population and urbanization, the demand for efficient resource management is increasing.
- Traditional systems often depend on manual effort, leading to wastage of time, water, energy, and manpower.
- There is a need for smart, automated, and data-driven solutions to improve efficiency and reduce human intervention.
- Advancements in IoT, sensors, and connectivity make it possible to monitor and control systems in real time.
- This project is motivated by the vision of building a sustainable, cost-effective, and intelligent solution that can benefit both agriculture and other real-world applications.

Features

- Real-Time Sensor Integration: Simultaneously tracks pH, TDS, Temperature, and Humidity levels.
- Wireless Data Sync: Uses an IoT framework to send sensor readings directly from the device to the cloud.
- Predictive Regression Analysis: Uses a regression model to turn raw data into future water quality trends.
- Disease Risk Scoring: Calculates a specific "Safety Index" to estimate the likelihood of water-borne diseases.
- Instant Alert System: Automatically triggers notifications whenever water parameters hit unsafe levels.
- Interactive Dashboard: A live visual interface for monitoring current stats and reviewing past data.

Architecture

Sensors

1

Data Collection

Sensors gather environmental or operational data

Initial Processing

ESP32 receives and preprocesses sensor data

2

ESP32

Cloud/Server

3

Data Analysis

Data is sent to the cloud for regression modeling

Risk Assessment

Model analyzes data to identify potential risks

4

Regression Model

Risk Output

5

Risk Indication

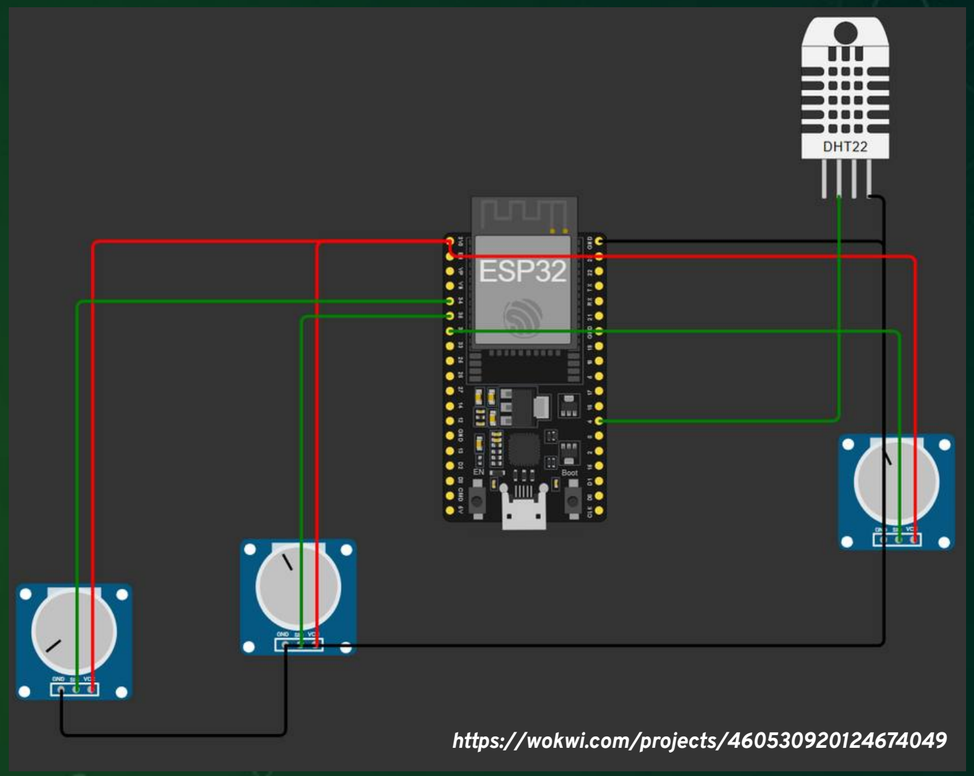
Model generates a risk score or assessment

Decision Support

Users view risk information and make informed decisions

6

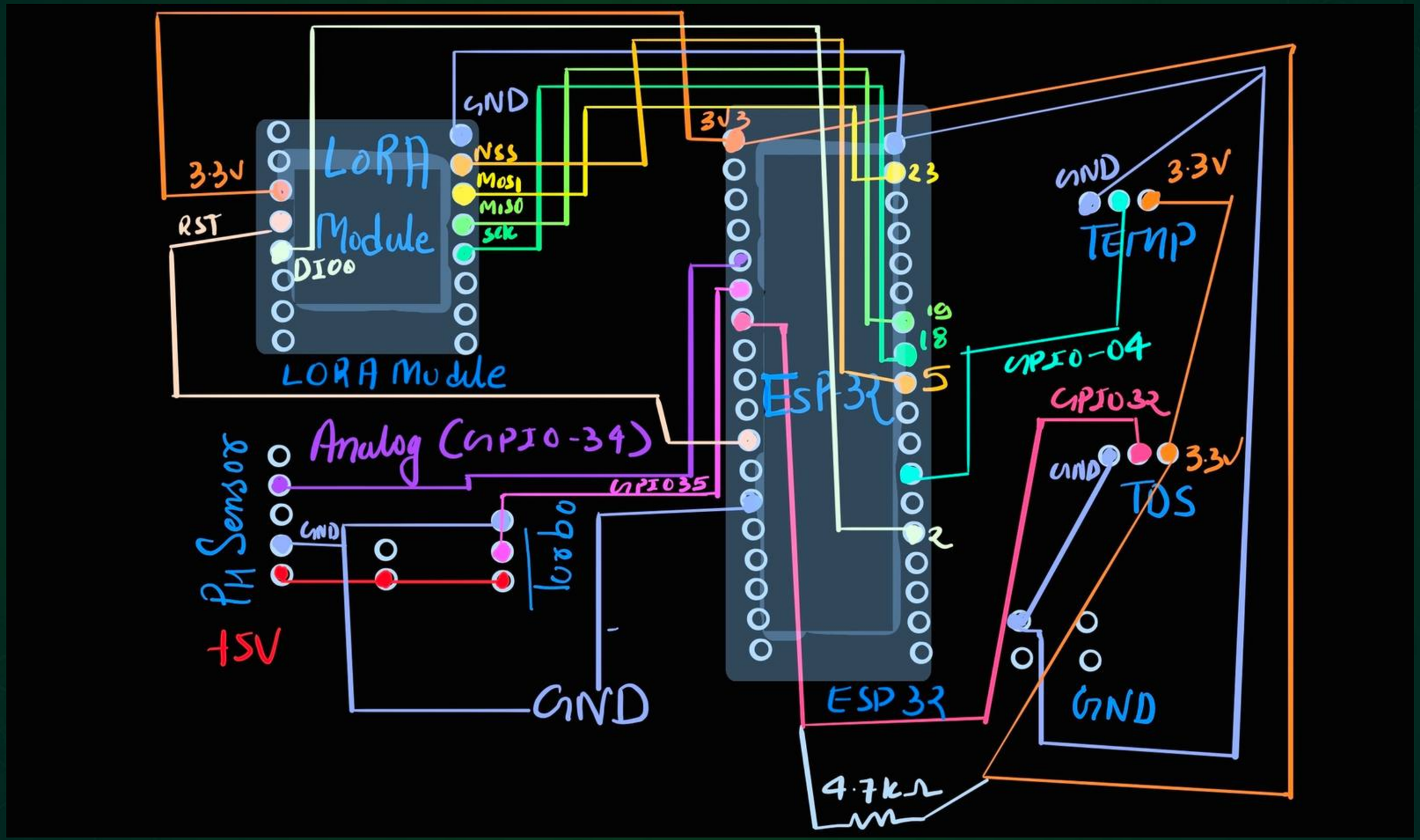
User Interface



List Of Components

ESP-32, pH Sensor , LoRa Module , TDS Sensor, Turbidity Sensor, Temperature Sensor

Circuit Diagram



Methodology

- ESP32 nodes collect data from pH, TDS, turbidity, and temperature sensors
- Data is transmitted through a LoRa mesh network to a central gateway
- Gateway sends data to a FastAPI backend server
- Data is stored in MongoDB databases.
- Data is cleaned and processed for analysis
- Classifier model classifies water as safe or unsafe
- LSTM models predict future water quality trends
- A risk score (Water Safety Index) is calculated
- React + Tailwind dashboard displays real-time and historical data
- Alerts are triggered when unsafe conditions are detected

Expected Outcome

- Real-time monitoring of water quality parameters (pH, TDS, Turbidity, Temperature) using IoT sensors.
- Continuous transmission of sensor data to a cloud platform for storage and analysis.
- Visualization of water quality data through graphs and dashboards for easy monitoring.
- Prediction of water safety level using a regression-based model.
- Early detection of abnormal water conditions that may indicate contamination.
- Alert generation when water quality becomes unsafe.

References

- Verma, S., & Gupta, R. (2023). IoT-based water quality monitoring system using ESP32. International Journal for Research in Applied Science & Engineering Technology (IJRASET). https://www.ijraset.com/research-paper/iot-based-water-quality-monitoring-system-using-esp32
- Hasan, T. F., Kabashi, N. A., Alam, M. Z., & Wahab, M. F. (2024). Water quality monitoring using machine learning and IoT: A review. Chemical and Natural Resources Engineering Journal. https://www.researchgate.net/publication/387545436
- ABoccadoro, P., et al. (2020). Water quality prediction on an IoT device using LSTM. IEEE/IoT-based system (WaterS project).
- https://arxiv.org/abs/2007.13436
- Sharma, N., & Kaur, P. (2025). Intelligent IoT-based water quality monitoring and predictive analysis. AISE Journal, 3(1), 1–10. https://aise.cultechpub.com/index.php/aise/article/view/16
- Mahesh, N. (2024). Water quality prediction using LSTM with combined normalization techniques. Journal of King Saud University – Computer and Information Sciences. https://www.sciencedirect.com/science/article/pii/S194439862400211X